

PENETRATION TESTING REPORT

Footprinting & Network Scanning Phases

W2-PM-FINAL | Cybersecurity | NetworkWalks

Field	Detail
Pentester name	Abdullatif Abuzannad (Cybersecurity Intern)
Program / Batch	B082 — NetworkWalks Cybersecurity Internship
Date	19 August 2026
Modules completed	W2-PM1 (Footprinting with multiple Kali tools) W2-PM5 (Network scanning with Zenmap)
Target 1	networkwalks.com (written permission secured)
Target 2	My own local home network (owned by me)
Permission secured?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I performed all the activities in this report only against systems I own or systems I had written permission to test. Everything here is for education and learning only. None of it should be used to break the law.

The instructor, the authors and NetworkWalks are not responsible for how this knowledge is used — every action a person takes is their own responsibility. In most countries, accessing a system without permission is a crime even if nothing is damaged.

2. Introduction

This report covers the Week 2 part of my Cybersecurity internship at NetworkWalks. It has two activities. In the first, I footprinted the networkwalks.com website using several Kali Linux tools (module W2-PM1). In the second, I scanned my own home network using Zenmap to find the live devices on it (module W2-PM5).

Put together, the two activities show the first two phases of a penetration test: gathering public information about a target from the outside (footprinting), and then mapping the live hosts on a network from the inside (scanning). One phase leads into the other — you cannot plan an attack on something you have not first understood.

For each step below I have included the command I ran, what I saw in the result, a screenshot as proof, and a short note on why the finding matters. All the footprinting was done in Kali Linux, and the scanning was done on a Windows PC with Zenmap installed.

3. Tools Used

The table below lists each tool I used and what it was for.

Tool	Purpose
whois	Find the domain registration details (owner, dates, name servers).
whatweb	Fingerprint the web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan my local network to find live hosts, IPs and MAC addresses.
Windows CMD	Find my own local IP and MAC address (ipconfig).

4. Activities Performed

4.1 Footprinting & Reconnaissance

I ran reconnaissance against networkwalks.com using Kali Linux tools. Each tool gathered a different piece of public information, and together they built a full picture of the target.

whois — I started with the domain registration record. It showed the domain is registered with GoDaddy, created in 2019 and valid until 2027. The owner's personal details are hidden behind a privacy service (Domains By Proxy), but the name servers still pointed to HostGator, so I could tell straight away where the site is hosted.

nslookup — I then resolved the domain name to its real IP address, which came back as 192.232.216.135. This is the actual machine the website runs on, and it is the anchor for everything else.

curl -I — Reading the HTTP response headers gave up a lot without loading the page. The server is Apache with an Nginx cache, and the headers confirmed the site runs WordPress. A cookie named __wpdm_client revealed the WP Download Manager plugin, and a link header exposed the WordPress REST API at /wp-json/.

wafw00f — I checked whether a firewall was protecting the site. It reported that the site sits behind ModSecurity (SpiderLabs). This is actually a good sign for the site owner, because it means simple attacks would be blocked or logged.

dnsrecon — Finally I enumerated the DNS records. This gave the fullest picture: the HostGator name servers again, the DNS software version (BIND 9.16.23), the mail server, an SPF record listing who can send mail for the domain, and cPanel service records that reveal the hosting control panel.

whatweb (could not run) — I was not able to run whatweb. It timed out on the Kali VM, and on a second machine (Ubuntu) it failed to launch with a library error, which means the installation was broken rather than the network. Instead of spending more time on it, I noted the issue and moved on. Importantly, this left no real gap: the same web technology information (WordPress, the WP Download Manager plugin, Apache, and the server IP) had already been found through the HTTP headers and the DNS lookups. Only the exact version numbers were missed. This taught me that reconnaissance does not rely on a single tool — the same information usually leaks through more than one channel.

4.2 Network Scanning with Zenmap

For the second activity I used Zenmap to discover the live devices on my own home network. Zenmap is no longer available on modern Kali Linux (it depends on old Python 2 libraries), so I installed the official Nmap and Zenmap package on Windows and scanned my own network from there.

First I used ipconfig to find my local IP address (192.168.18.9) and subnet mask (255.255.255.0), which gave me a /24 network. I entered 192.168.18.0/24 into Zenmap and ran a Ping scan (nmap -sn), which finds which hosts are up without scanning any ports.

The scan found 5 live hosts, including my own laptop. The table below lists them with their IP and MAC addresses.

IP address	MAC address	Device
192.168.18.1	40:48:6E:A2:0C:00 (Nokia)	Router / gateway (dsldevice.lan)

IP address	MAC address	Device
192.168.18.8	06:1B:C8:E1:AA:B3 (unknown)	Device with a randomized MAC
192.168.18.9	90:E8:68:2B:52:03 (MediaTek)	My laptop (from ipconfig /all)
192.168.18.41	22:70:4B:90:20:AF (unknown)	Device with a randomized MAC
192.168.18.96	9A:3B:61:D4:7B:42 (unknown)	Device with a randomized MAC

An interesting finding here is that three of the devices have randomized MAC addresses. This is a privacy feature that modern phones and laptops use to avoid being tracked, and it is why Nmap could not match them to a manufacturer. My own laptop showed as up but with no MAC in the scan, because a device does not ARP itself — so I found its MAC separately using ipconfig /all. This is different from a clean lab network, where every device is usually easy to identify, and it shows how a real network looks.

Note on addresses: all the IP and MAC addresses shown are from my own private home network. Private addresses (192.168.x.x) and device MACs only mean anything inside that local network.

5. Risk Analysis / Impact

The table below lists what I observed and the possible impact of each finding. These are observations from information gathering only — they are not confirmed vulnerabilities.

#	Finding	Evidence	Possible impact	Risk
1	Web technology exposed	curl/headers showed WordPress + WP Download Manager	Version info could be looked up for known issues	Medium
2	Server IP identifiable	nslookup resolved to 192.232.216.135	Shows the network location of the web service	Low
3	HTTP info exposed	curl exposed /wp-json/ REST API	Helps fingerprint the stack and find entry points	Low
4	WAF identifiable	wafw00f found ModSecurity	Reveals the site's security setup	Low
5	DNS info exposed	dnsrecon found NS, MX, SPF, cPanel records	Helps build a broader picture of the infrastructure	Medium
6	Multiple live hosts on LAN	Zenmap found 5 live hosts	Unknown devices may be present on a network	Medium

Risk key: Medium / Low. The exercises were information gathering and host discovery only. No exploitation was done, so finding a version, an IP or a DNS record does not by itself mean a system is vulnerable — further authorized testing would be needed to confirm any real issue.

6. Recommendations

Based on what I observed, I would suggest the following:

1. Review what technology information is publicly visible (CMS, plugins, versions).
2. Keep the CMS, plugins and other software updated against current security advisories.
3. Review HTTP response headers to avoid exposing unnecessary technical detail.
4. Check DNS records regularly so only the required records and services are public.
5. Keep the WAF (ModSecurity) enabled and tuned, since it already blocks simple attacks.
6. Scan your own network from time to time to see which devices are active.
7. Investigate any unknown or unexpected device found on the network.
8. Only perform reconnaissance and scanning where you have proper authorization.

7. Conclusion

During Week 2 of my internship I completed practical activities covering footprinting, reconnaissance and network scanning.

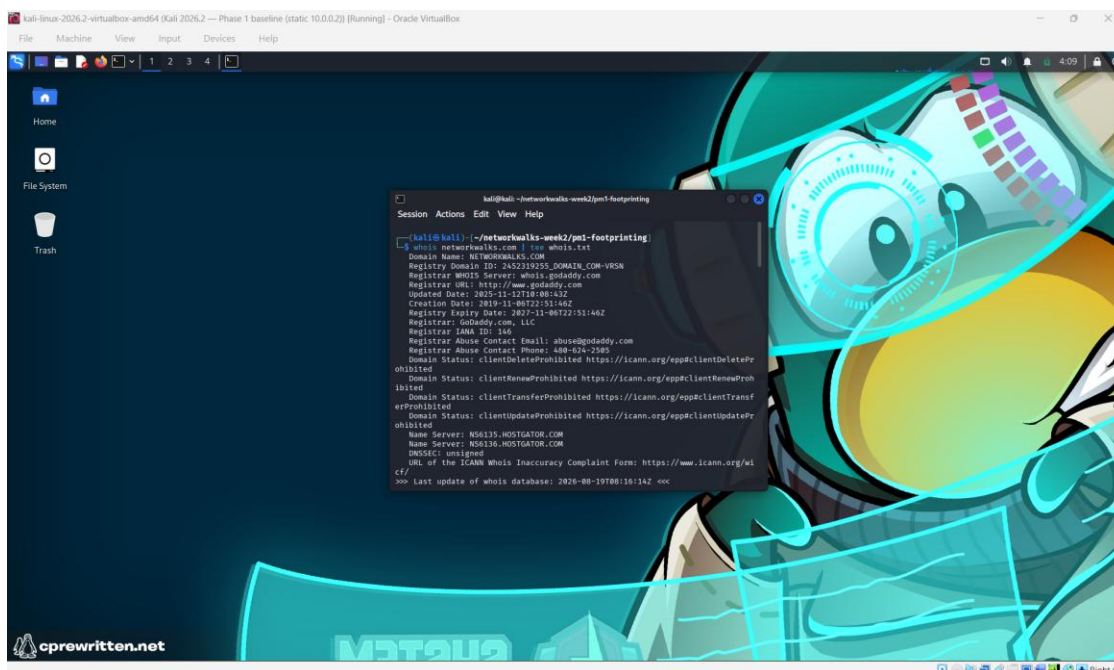
In the footprinting activity I used Kali Linux tools to collect public information about a target domain. I learned how whois shows domain and hosting details, nslookup resolves a name to an IP, curl reveals the technology through HTTP headers, wafw00f detects a firewall, and dnsrecon maps out the DNS records. When whatweb would not run, I was still able to confirm the same information through the other tools, which showed me that recon does not depend on any single tool.

In the scanning activity I used Zenmap on my own network to find the live devices, their IP and MAC addresses, and to save a network topology. I found that some devices hide their identity using MAC randomization, which is something a real network shows but a lab network usually does not.

Overall, these exercises showed me how much can be learned about a target before any attack is attempted, just by reading public information and network responses. They also showed me the importance of writing findings up clearly, and of only ever working within an authorized scope.

8. Evidence Collected

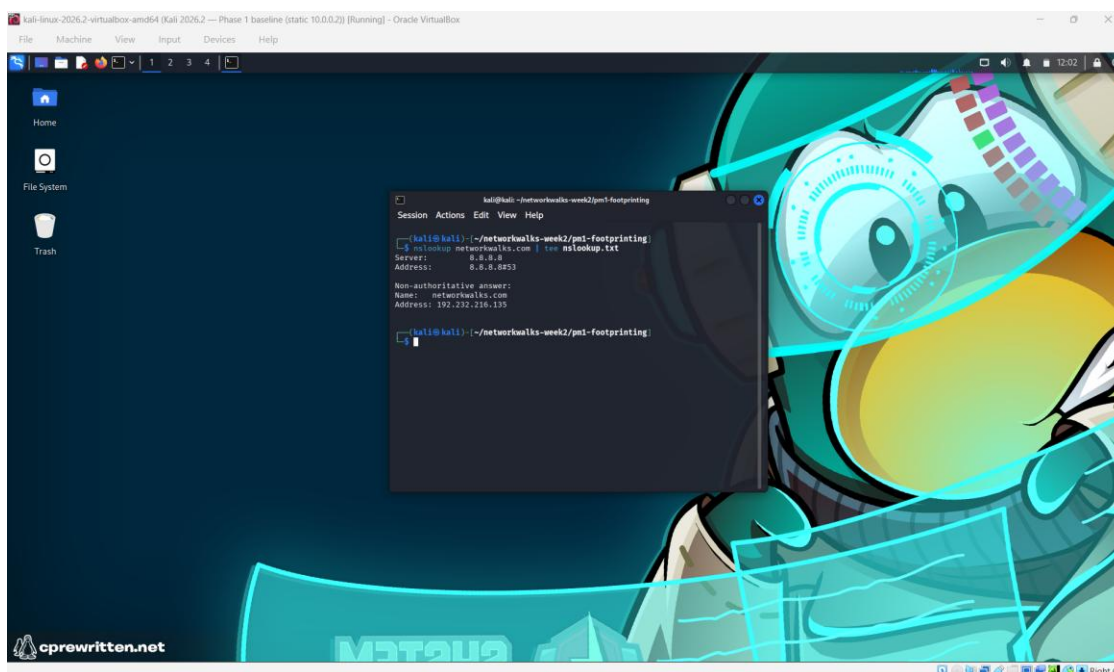
Footprinting — whois (domain registration):



```
kali@kali:~/networkwalks-week2/pm1-footprinting$ whois networkwalks.com | tee whois.txt
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452519255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T18:08:43Z
Creation Date: 2019-11-06T22:53:16Z
Registry Expiry Date: 2027-11-06T22:53:16Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2895
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/whois-inaccuracy-complaint-form
Last update of whois database: 2026-08-19T08:16:14Z
```

whois networkwalks.com — GoDaddy registrar, HostGator name servers

Footprinting — nslookup (resolve to IP):

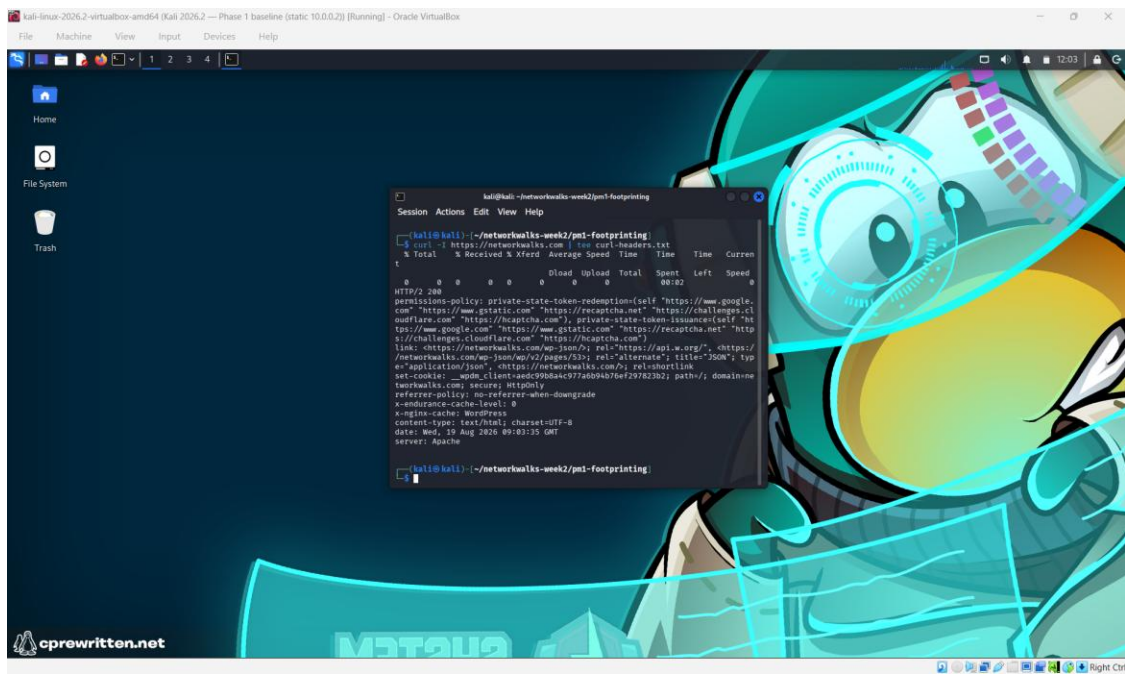


```
kali@kali:~/networkwalks-week2/pm1-footprinting$ nslookup networkwalks.com | tee nslookup.txt
Server: 8.8.8.8
Address: 8.8.8.853

Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
```

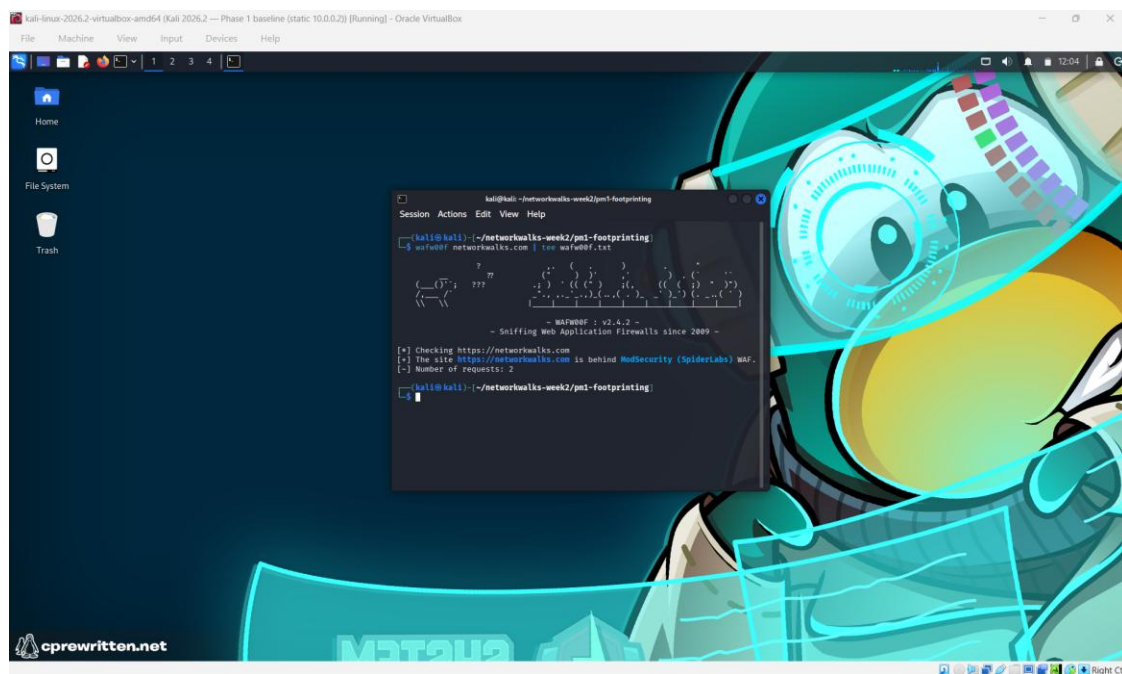
nslookup networkwalks.com — resolves to 192.232.216.135

Footprinting — curl -I (HTTP headers):



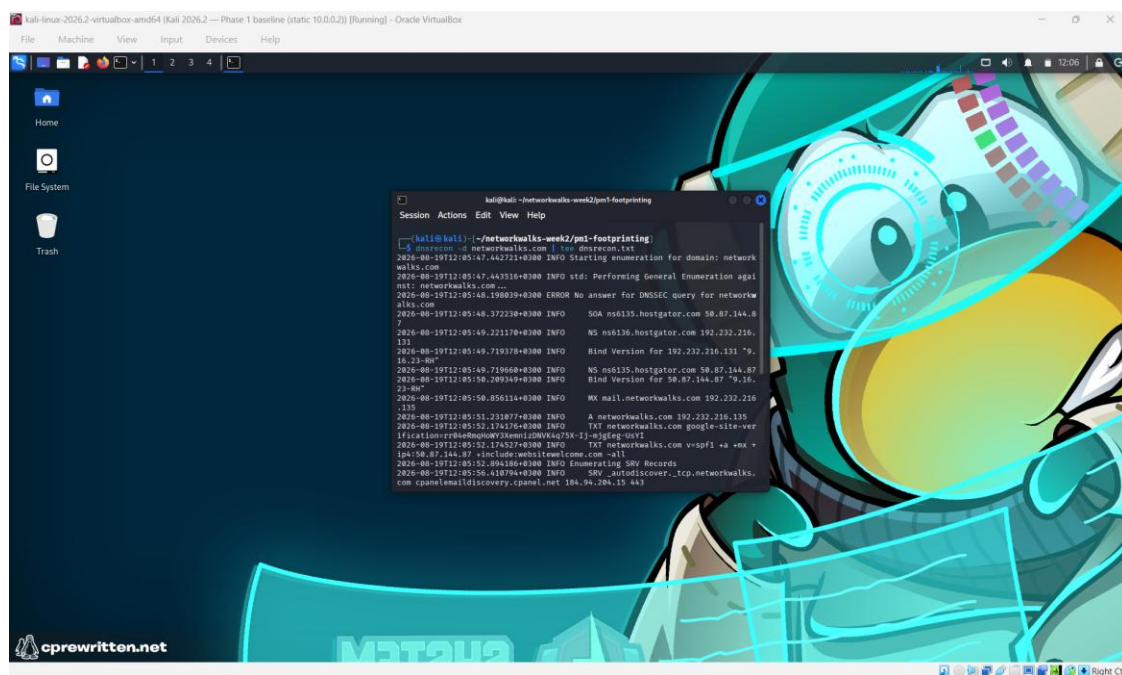
curl -I — Apache, WordPress, WP Download Manager cookie, /wp-json/ API

Footprinting — wafw00f (firewall detection):



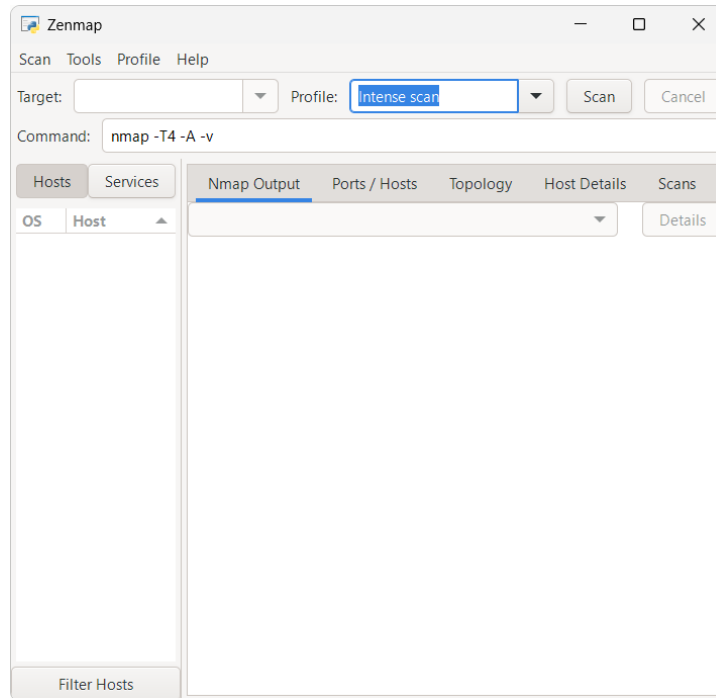
wafw00f — site is behind ModSecurity (SpiderLabs)

Footprinting — dnsrecon (DNS records):



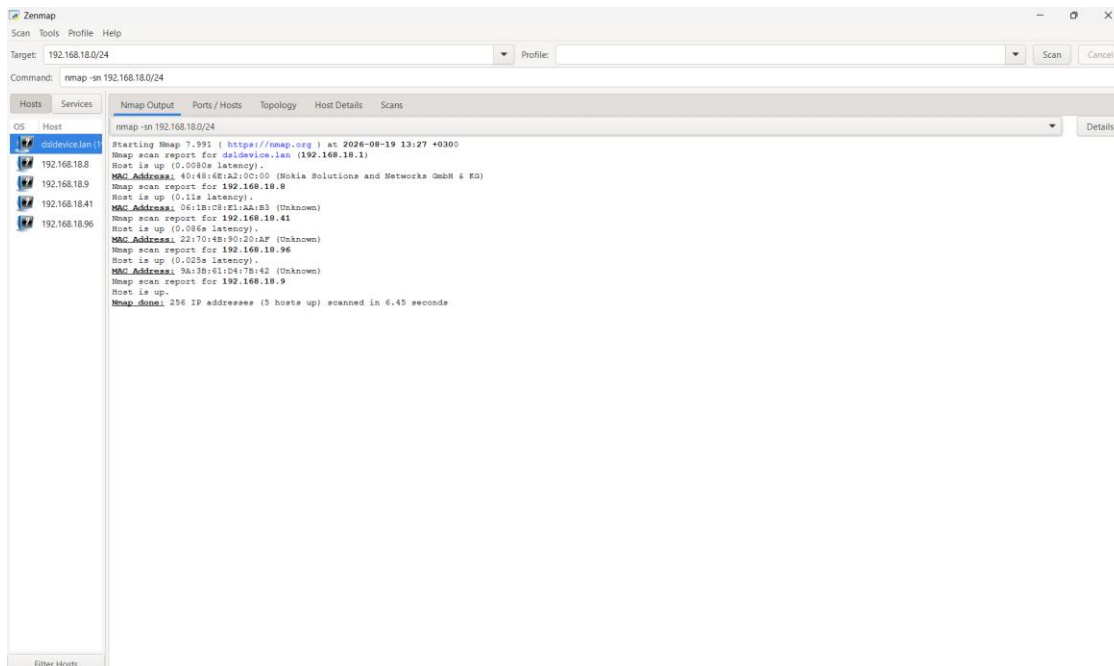
dnsrecon -d networkwalks.com — NS, BIND version, MX, SPF, cPanel records

Scanning — Zenmap installed on Windows:



Zenmap running on Windows

Scanning — Nmap output (live hosts and MACs):



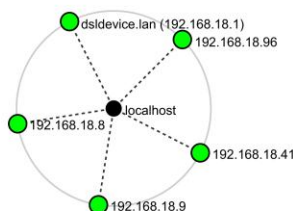
nmap -sn 192.168.18.0/24 — 5 hosts up with MAC addresses

Scanning — local IP and MAC (ipconfig, filtered):

```
Command Prompt
C:\Users\abuza>ipconfig /all | findstr /i "Wireless Physical IPv4 Subnet"
Physical Address. . . . . : 0A-00-27-00-00-0B
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Wireless LAN adapter Local Area Connection* 1:
Physical Address. . . . . : 92-E8-68-2B-52-23
Wireless LAN adapter Local Area Connection* 2:
Physical Address. . . . . : 92-E8-68-2B-52-33
Wireless LAN adapter Wi-Fi:
Description . . . . . : MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
Physical Address. . . . . : 90-E8-68-2B-52-03
IPv4 Address. . . . . : 192.168.18.9(Preferred)
Subnet Mask . . . . . : 255.255.255.0
C:\Users\abuza>
```

ipconfig /all (filtered) — my laptop 192.168.18.9, MAC 90:E8:68:2B:52:03

Scanning — network topology:



Zenmap topology of 192.168.18.0/24 — saved as PDF

Author

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Repository: <https://github.com/abdullatif13-ab/abdullatif13-ab-networkwalks-B082-week2-Footprinting-Network-Scanning>

Program: Cybersecurity program at NetworkWalks | Week 2 | Modules W2-PM1 + W2-PM5